

torch.nn.functional.kl_div#

https://pytorch.org/docs/stable/generated/torch.nn.functional.kl_div.html

Description:

Compute the KL Divergence loss. Refer - The Kullback-Leibler divergence Loss See `KLDivLoss` for details. `input` (Tensor) – Tensor of arbitrary shape in log-probabilities. `target` (Tensor) – Tensor of the same shape as `input`. See `log_target` for the target's interpretation.

Function Signature

```
torch.nn.functional.kl_div(input, target, size_average=None,
                             reduce=None, reduction='mean', log_target=False)[source]#
```

Parameters

Return type

Returns:

Tensor

Notes & Warnings

NOTE

Note `size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction.

⚠ WARNING

Warning `reduction = 'mean'` doesn't return the true kl divergence value, please use `reduction = 'batchmean'` which aligns with KL math definition.

Detailed Documentation

Documentation

Compute the KL Divergence loss.

Refer - The Kullback-Leibler divergence Loss

See `KLDivLoss` for details.

`input` (Tensor) – Tensor of arbitrary shape in log-probabilities.

`target` (Tensor) – Tensor of the same shape as `input`. See `log_target` for the target's interpretation.

`size_average` (bool, optional) – Deprecated (see `reduction`).

`reduce` (bool, optional) – Deprecated (see `reduction`).

`reduction` (str, optional) – Specifies the reduction to apply to the output: 'none' | 'batchmean' | 'sum' | 'mean'. 'none': no reduction will be applied 'batchmean': the sum of the output will be divided by the batchsize 'sum': the output will be summed 'mean': the output will be divided by the number of elements in the output Default: 'mean'

`log_target` (bool) – A flag indicating whether target is passed in the log space. It is recommended to pass certain distributions (like softmax) in the log space to avoid numerical issues caused by explicit log. Default: False

`size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override `reduction`.

`reduction = 'mean'` doesn't return the true kl divergence value, please use `reduction = 'batchmean'` which aligns with KL math definition.